

# Rakthesh TG

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## Summary

Computer Science undergraduate (CGPA 9.36) at VIT Chennai with full-stack development experience and growing expertise in agentic AI systems (LangGraph, RAG). Has shipped a deployed production web application end-to-end and built multi-agent AI workflows with retrieval and human-in-the-loop controls. Seeking a Software Engineering or AI/ML internship.

## Education

**Vellore Institute of Technology, Chennai**

Aug 2024 – May 2028

*B.Tech in Computer Science and Engineering — CGPA: 9.36/10*

- Cleared JEE Advanced 2024; JEE Mains 2024: 97.07 percentile among 1.2M+ candidates
- Relevant coursework: Data Structures & Algorithms, DBMS, Operating Systems, Computer Networks, OOP

## Technical Skills

**Languages:** C, C++, JavaScript, Python, SQL, PL/SQL

**Web/Full-Stack:** React, Express.js, MySQL, REST APIs, HTML/CSS

**AI/Agentic:** LangGraph, RAG pipelines (Cohere, Pinecone, Groq), Whisper, agentic workflows

**CS Fundamentals:** Data Structures & Algorithms, OOP, DBMS, Operating Systems, Computer Networks

**Tools:** Git, VS Code, Oracle SQL, NS-3, Vercel

## Projects

**UniFocus — Full-Stack Productivity Suite** | *React, Express.js, MySQL* 2026

- Built and deployed UniFocus, a full-stack productivity suite (React, Express.js, MySQL on Aiven Cloud) covering habit tracking, to-do lists, a calendar, timetable, stress logging, and a Pomodoro timer, live at uni-focus-eight.vercel.app
- Designed a REST API with 28 endpoints across 8 resource domains (auth, habits, todos, calendar, timetable, pomodoro, stress, preferences), with bcrypt-hashed password storage and connection-pooled MySQL access
- Built a custom SVG sine-wave liquid-fill animation for the Pomodoro timer using per-bubble keyframe injection, and a contact section with staggered, physics-inspired magnetic-pull entrance animations

**Multi-Agent Customer Support System** | *LangGraph, RAG, SQLite* 2026

- Designed a multi-agent customer support automation system using LangGraph, implementing a stateful graph with conditional routing across Sales, Technical, Billing, and Account support agents
- Integrated a RAG pipeline over company policy, pricing, and technical documentation to ground agent responses in source material
- Implemented SQLite-backed conversational memory enabling agents to recall prior customer interactions across sessions

**PaperTrail — RAG-Based Paper Explainer** | *React, TanStack Start* In Progress

- Building PaperTrail (React, TanStack Start, TypeScript), a tool that turns a pasted paper URL into a structured summary, an interactive concept-map visualization, and a contextual Q&A panel (MVP stage, pre-deployment)
- Built a RAG pipeline: PDF text extraction and chunking, Cohere embeddings, Pinecone vector storage per-paper namespace, and Groq-served LLM generation grounded in retrieved chunks
- Designed the output as three structured lenses (summary, concept map, relevance) instead of a flat chat interface, targeting students who need fast comprehension during live lectures

## Academic Experience

**Database Management Systems & Computer Networks Labs**

2025

*Vellore Institute of Technology, Chennai*

- Designed relational schemas and executed complex SQL queries (joins, subqueries, views) in Oracle SQL, applying normalization and integrity constraints
- Simulated Point-to-Point, Wi-Fi, and Bluetooth network topologies in NS-3; diagnosed and resolved a Bluetooth IPv6/6LoWPAN misconfiguration causing 100% packet loss